Image Processing and Medical Imaging

Mathematical Morphology

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Basic morphological concepts

- Mathematical morphology was developed by Matheron and Serra.
- Morphological operations are used for the following purposes:
  - Image pre-processing (noise filtering, shape simplification).
  - Enhancing object structure (skeletonizing, thinning, thickening, etc.).
  - Segmentation.
  - Quantitative description of objects (area, perimeter, etc.).

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Morphological transformation

Definition. A **morphological transformation** $\Psi(X) \rightarrow X$, where is X is a set of points, is an operator with four properties.

- A **morphological operator** is composition of a morphological transformations followed by a measure which is a mapping to the set of real numbers.
- We will consider only the morphological transformation during this class.

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Four properties of morphological transformations

- Compatibility with translation.
- Compatibility with change of scale.
- Local knowledge.
- Upper semi-continuity.

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Compatibility with translation

Let the transformation Ψ depends on the positon of the origin O of the coordinate system, and denote such transformation by $\Psi_{O.}$ If all points are translated by the vector $-h$, it is expressed as Ψ_{-h}.

$$\Psi_O(X_h)=\Psi_{-h}(X))_{h.}$$

If Ψ does not depend on the position of the origin O, this principle reduces to

$$\Psi(X_h)=\Psi(X))_{h.}$$

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Compatibility with change of scale

Let λX represent the homothetic scaling of a point set X. Let Ψ_λ denote a transformation that depends on the positive parameter λ. Compatibility with change of scale is given by

$$\psi_\lambda(X)=\lambda\Psi((1/\lambda)X).$$

If Ψ does not depend on the scale λ then

$$\Psi(\lambda X)=\lambda\Psi(X).$$

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Local knowledge

The local knowledge principle consider the situation in which only a part of a larger structure can be examined. The local knowledge principle may be written symbolically as

$$(\Psi(X \cap Z)) \cap Z' = \Psi(X) \cap Z'$$

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Binary dilation

The morphological transformation **dilation** $\oplus$ combines two sets using vector addition. The dilation X $\oplus$B is the point set of all possible vector additions of pairs of elements, one from each of the set X and B

$$X \oplus B = \left\{ p \in Z^2 : p = x + b, x \in X \wedge b \in B \right\}$$

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Binary dilation

Binary image - X

0	0	0	0
0	1	1	0
0	0	0	0

Structuring element - B

0	1	0
1	1	1
0	1	0

Result image after dilation

0	1	1	0
1	1	1	1
0	1	1	0

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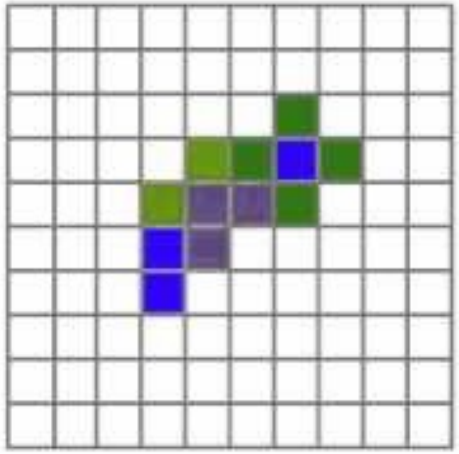
Example: binary dilation

Dilation

$$A \oplus B = \bigcup_{a \in A} (B)_a$$



Source: https://www.youtube.com/watch?time_continue=1&v=W5x-z63JFQY&feature=emb_logo

Binary erosion

The morphological transformation **erosion** combines two sets using vector addition. The erosion X $\ominus$ B is the point set determined by the following form:

$$X \Theta B = \left\{ p \in Z^2 : p = x + b \in X, \forall b \in B \right\}$$

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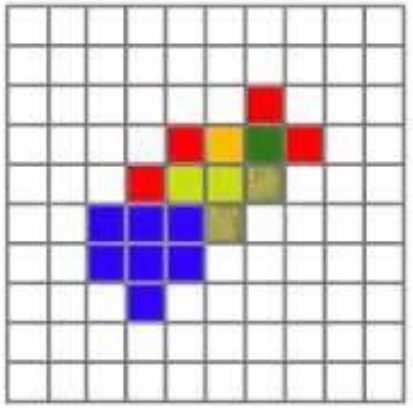
Example: binary erosion

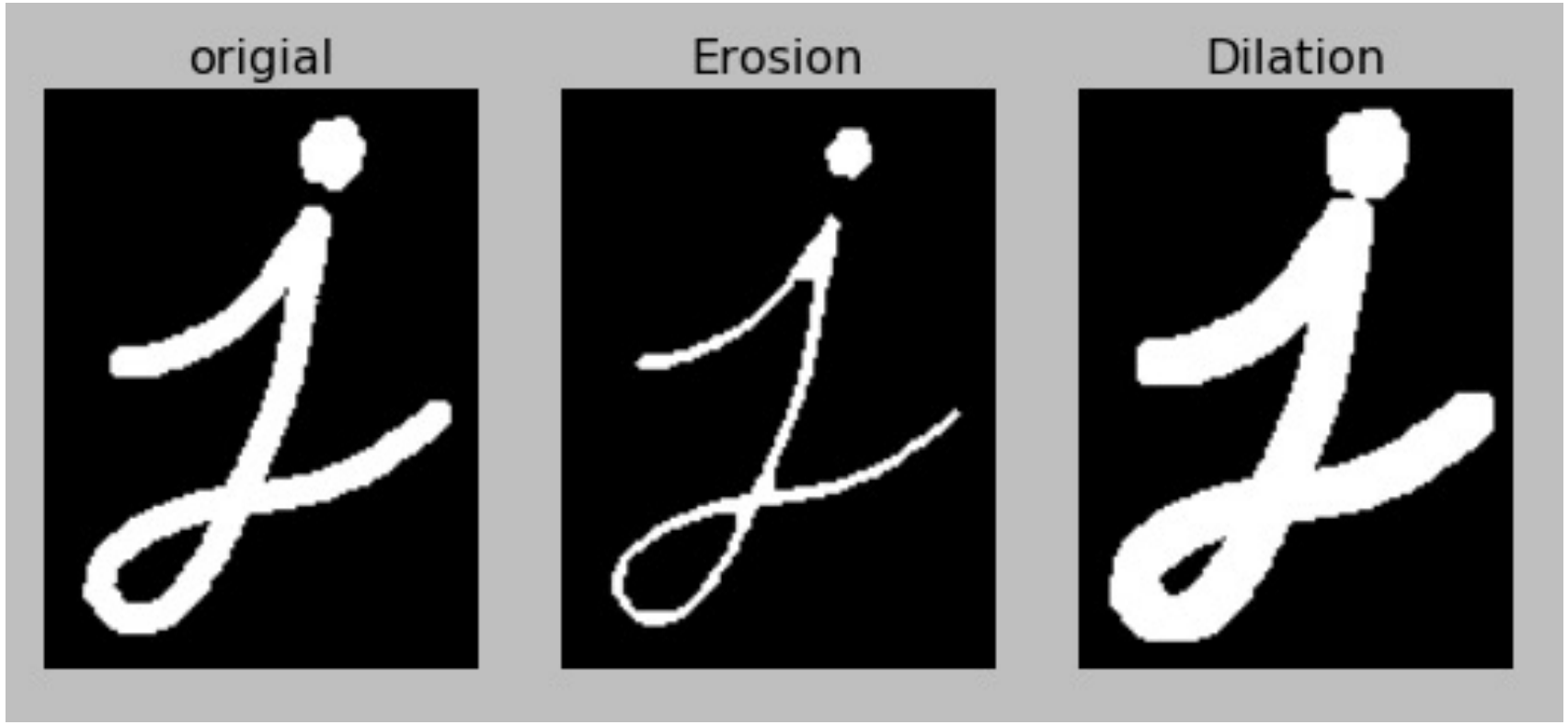
Erosion

$$A \ominus B = \{x : B_x \subseteq A\}$$



Source: https://www.youtube.com/watch?v=yLdjNi2rncI&feature=emb_logo

Example: binary dilation and erosion



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Source: https://docs.opencv.org/2.4/doc/tutorials/imgproc/erosion_dilatation/erosion_dilatation.html

Properties of dilation and erosion

- The dilation operation is **cummutative**, **associative**, **invariant to translation**, and **increasing transformation**.
- The erosion is **invariant to translation**, and **increasing transformation**.
- The erosion and dilation are **dual transformations**.

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Opening and closing

- Erosion and dilation are not inverse transformations. So we can make two new morphological transformations
- Erosion followed by dilation creates an important morphological transformation called **opening**. Notation: $X \circ B$
- Dilation followed by erosion is called **closing**. Notation: $X \bullet B$

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Opening and closing

- Opening and closing with an isotropic structuring element is used to eliminate specific image details smaller the structuring element.
- **Closing** connects objects that are close to each other, fills up small holes, and smoothes the object outline by filling up narrow gulfs.
- **Opening** removes small objects.
- Opening and closing are dual transformations.

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Example: opening and closing

Original Image | Opening | Closing

Source: https://towardsdatascience.com/computer-vision-for-beginners-part-2-29b3f9151874

Hit-or-miss transformation

- The **hit-or-miss transformation** is defined as

$$X \otimes B = \left\{ x : B_1 \subset X \wedge B_2 \subset X^C \right\}$$

- It can be defined as

$$X \otimes B = (X \Theta B_1) \cap (X^C \oplus B_2)$$

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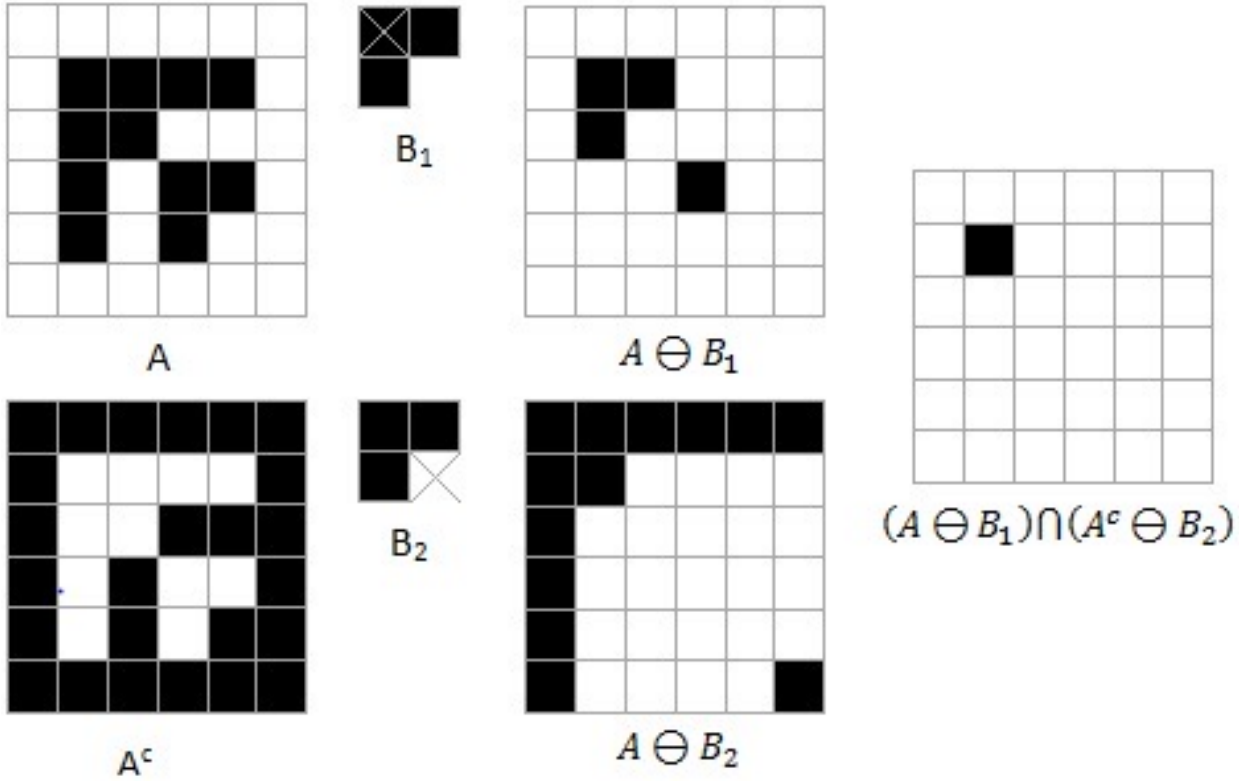
Example: Hit-or-miss transformation



Source: https://commons.wikimedia.org/wiki/File:Hitormisstransformatie.jpg

Gray-scale morphology

The gray-scale morpholgy based on using *sup* (dilation) and *inf* (erosion) instead of union and intersection. The dilation can be definied in this form:

$$(f \oplus b)(x) = \sup_{y \in B} \left[f(x - y) \right]$$

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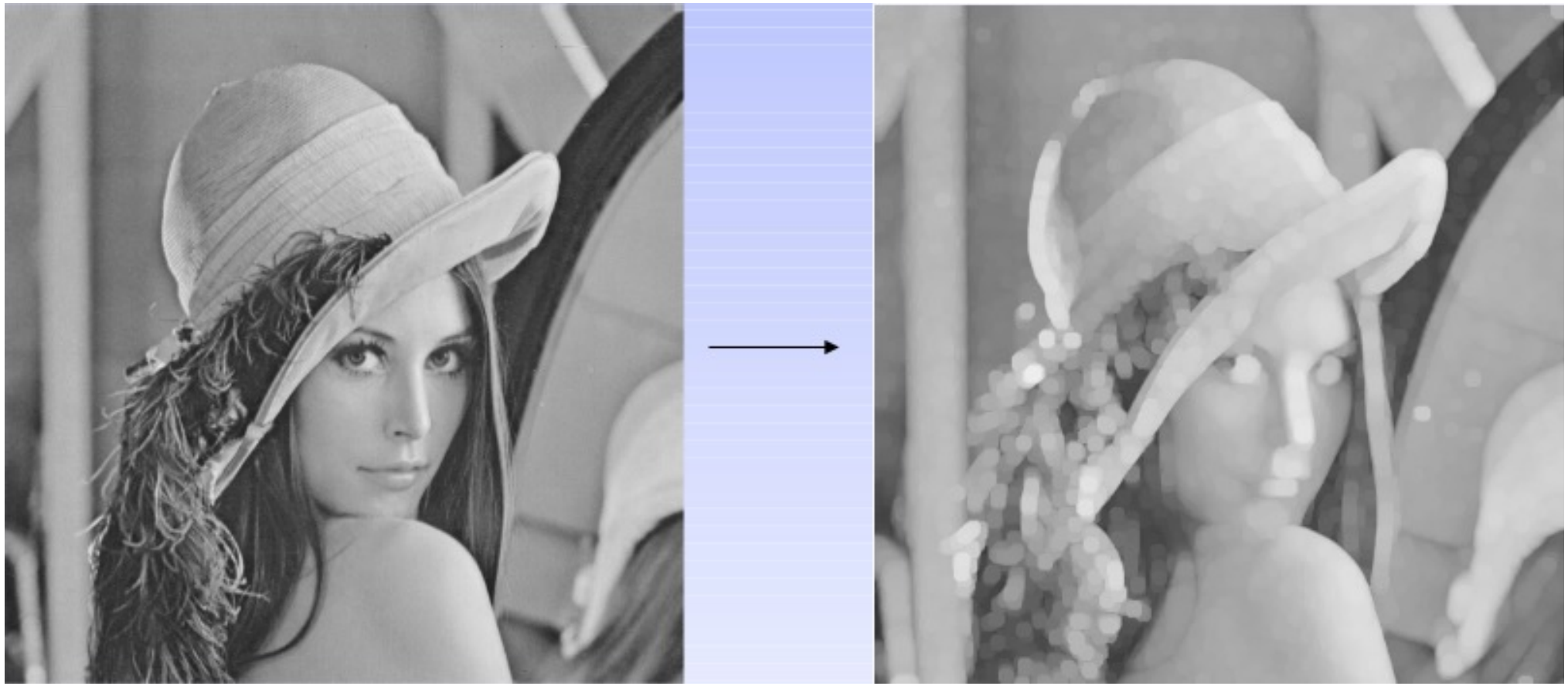
Gray-scale dilation



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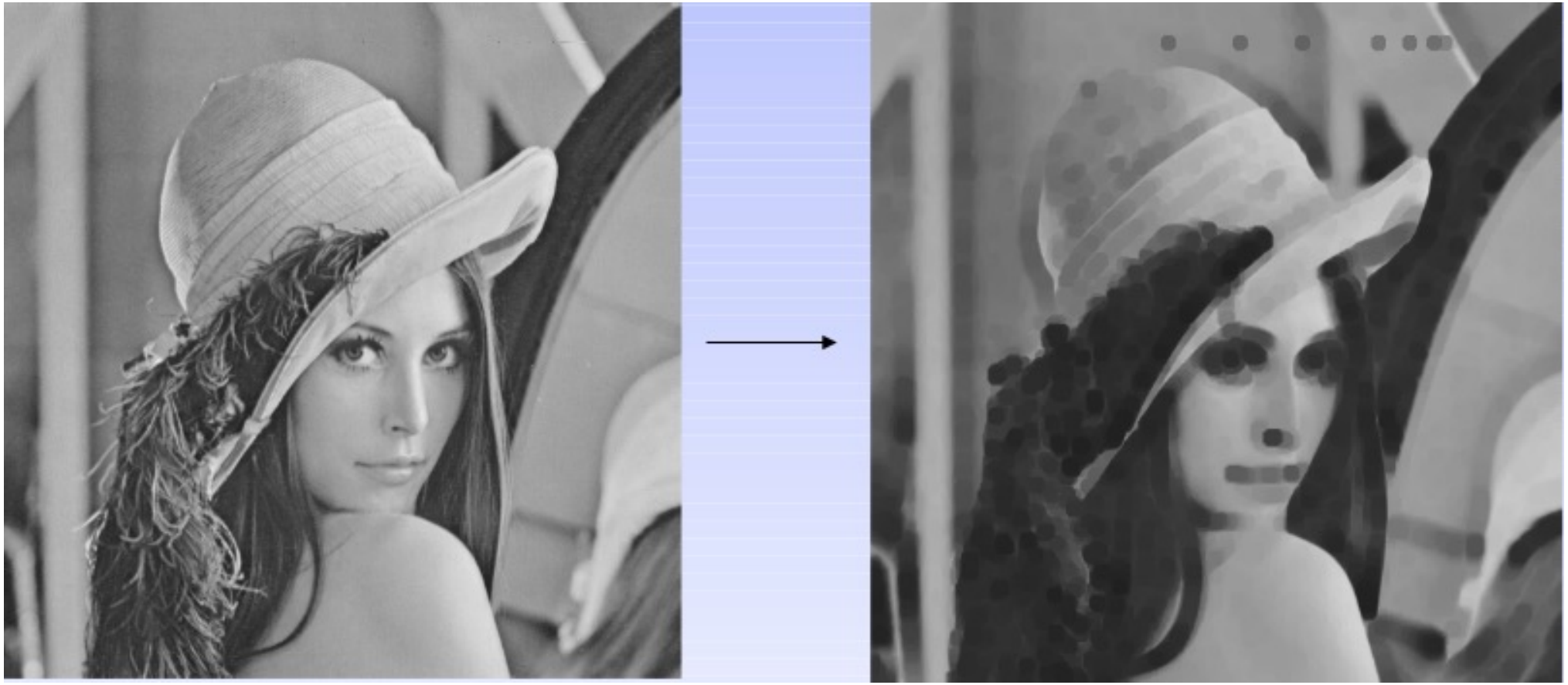
https://maverick.inria.fr/Membres/Adrien.Bousseau/morphology/morphomath.pdf

Gray-scale erosion



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https://maverick.inria.fr/Membres/Adrien.Bousseau/morphology/morphomath.pdf

More from mathematical morphology

- There are several morphological operators.
- Non-linear filtering (opening, closing) based on this technique.
- The effective execution is an important question.
- Hardware solutions for morphological operators.

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Thank you for your attention